

Soccer Kicking Accuracy: A Factorial Design with Blocking by Kicker

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Objective of the Study

Assess whether soccer ball size and initial kicking distance affect kicking accuracy. Accuracy was measured as the final distance between the ball and the target (yards).

This study was designed to answer the following questions:

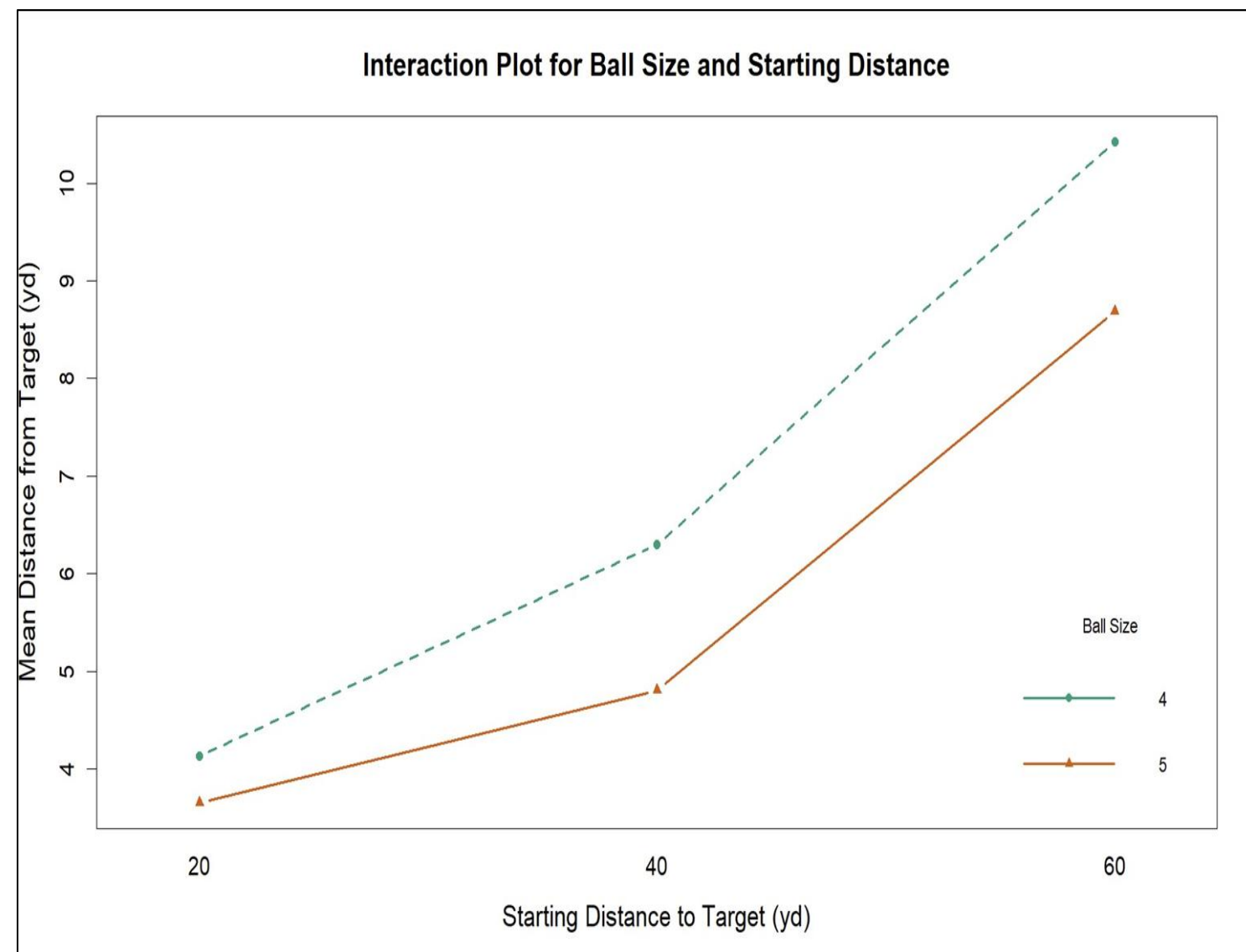
- ❖ Does soccer ball size affect kicking accuracy?
- ❖ Does the starting distance from the target affect kicking accuracy?
- ❖ Is there an interaction between ball size and starting distance?
- ❖ Does player experience influence the results enough to justify blocking?

Experimental Design

- ❖ 2 × 3 factorial design
- ❖ Blocking factor: **Kicker** (3 players)
- ❖ Factors and levels:
 - **Ball size:** Size 4, Size 5
 - **Initial distance:** 20 yards, 40 yards, 60 yards
- ❖ 5 replicates per treatment combination
- ❖ Randomized order
- ❖ Response variable:
 - **Final distance from the target**

Note: Smaller distance indicates higher accuracy

Interaction Plot



The interaction plot shows non-parallel trends between ball size and starting distance, suggesting a possible interaction effect.

Modeling and ANOVA Results

Step 1: Initial model

An initial model including the interaction term was fitted:

$$\text{Distance} \sim \text{Kicker} + \text{Ball Size} + \text{Starting Distance} + \text{Ball Size} \times \text{Starting Distance}$$

Step 2: Decision based on ANOVA

ANOVA results indicated that the interaction term was not statistically significant ($p = 0.704$), so it was removed.

Step 3: Final model

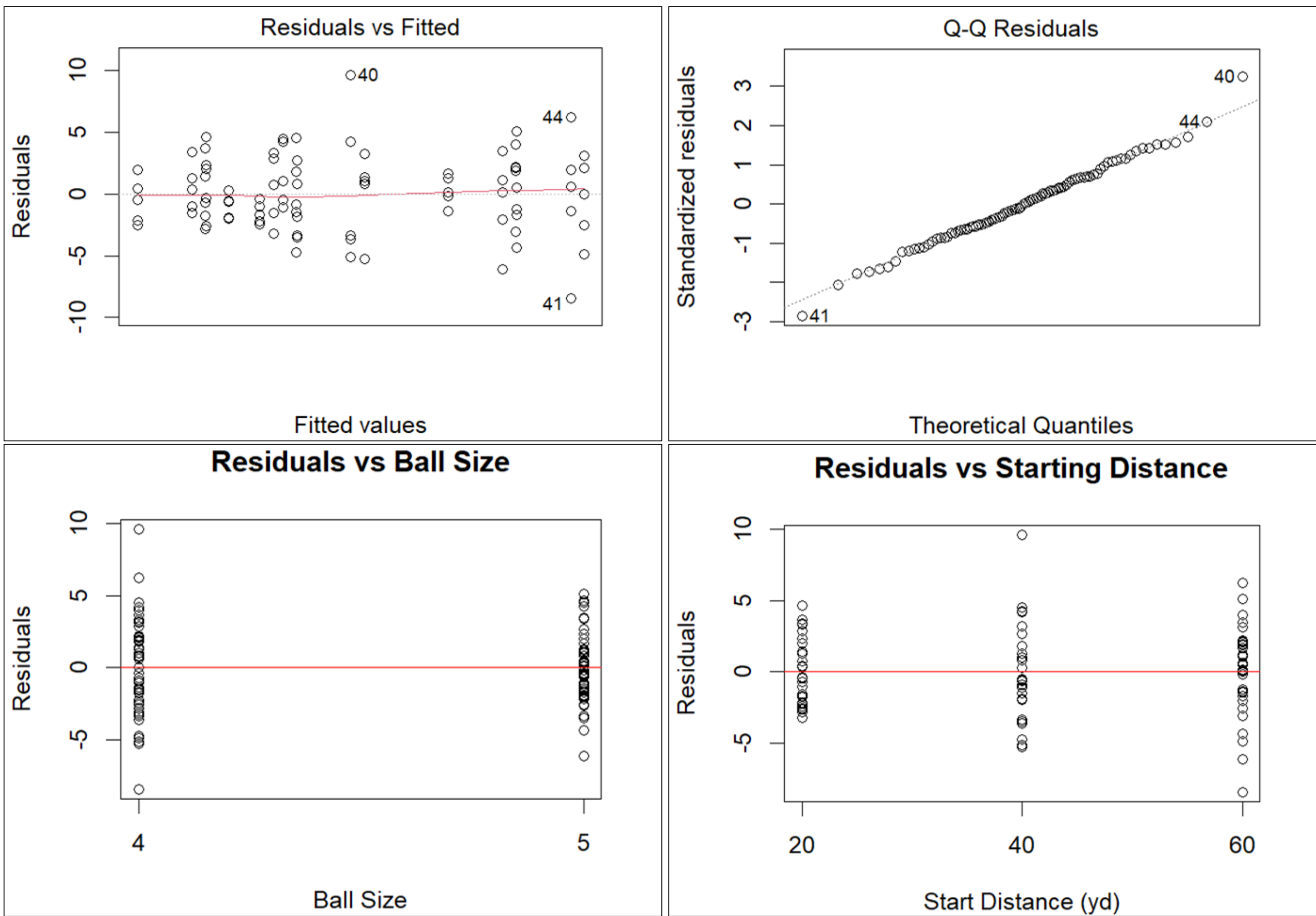
$$\text{Distance} \sim \text{Kicker} + \text{Ball Size} + \text{Starting Distance}$$

Key Results:

- ❖ Starting distance: significant ($p < 0.001$)
- ❖ Ball Size: marginal ($p \approx 0.059$)

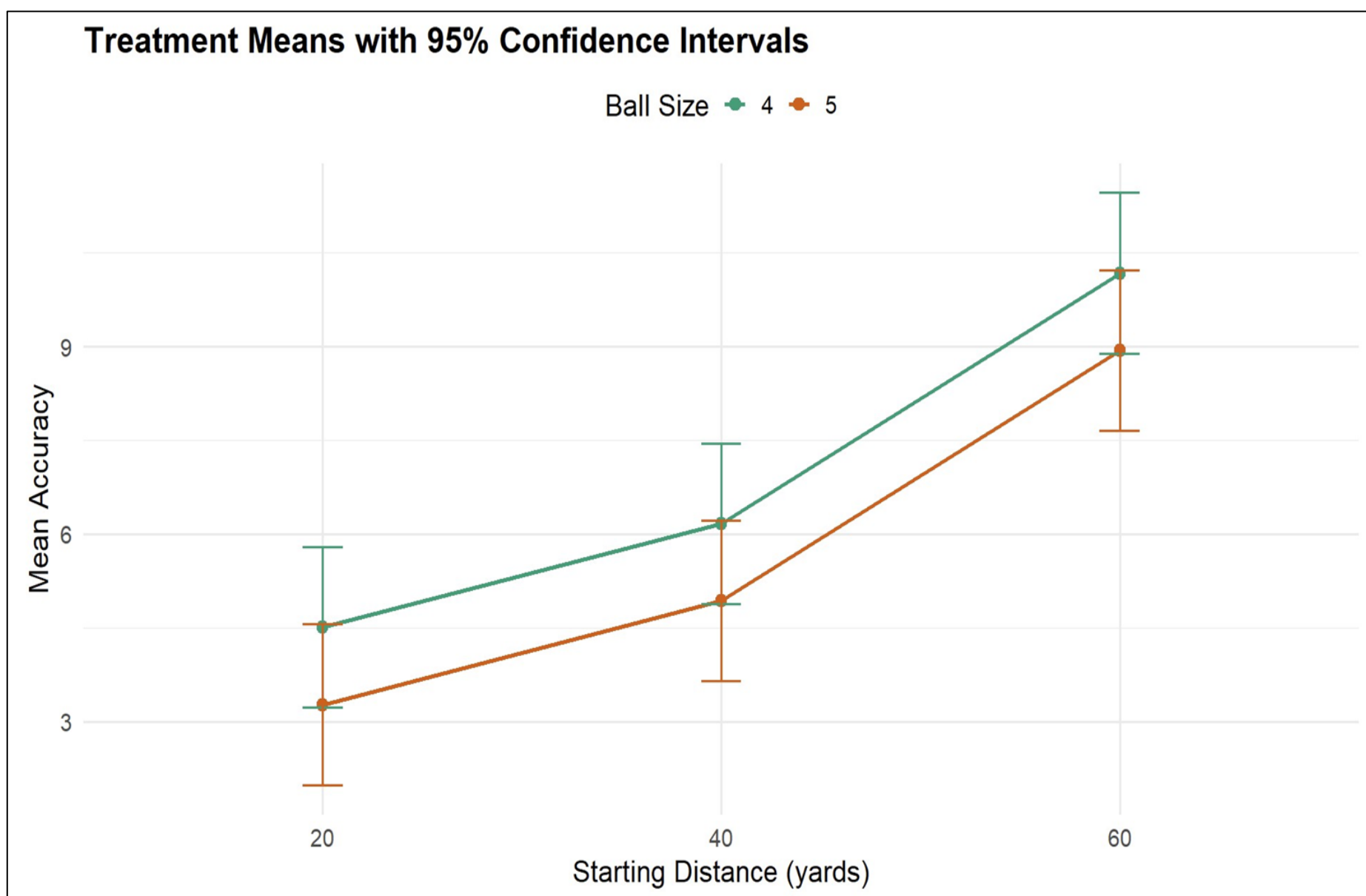
Step 4: Residual diagnostics

Residual diagnostics indicate that model assumptions are reasonably satisfied.



Treatment Means with 95% CIs

- ❖ Mean distance from the target increases as starting distance increases, indicating reduced accuracy at longer distances.
- ❖ For each starting distance, the confidence intervals for ball sizes 4 and 5 overlap, suggesting only a small difference in accuracy between ball sizes.
- ❖ The roughly parallel pattern of the lines indicates little to no interaction between ball size and starting distance, which is consistent with the ANOVA results.



Limitations and Sources of Variability

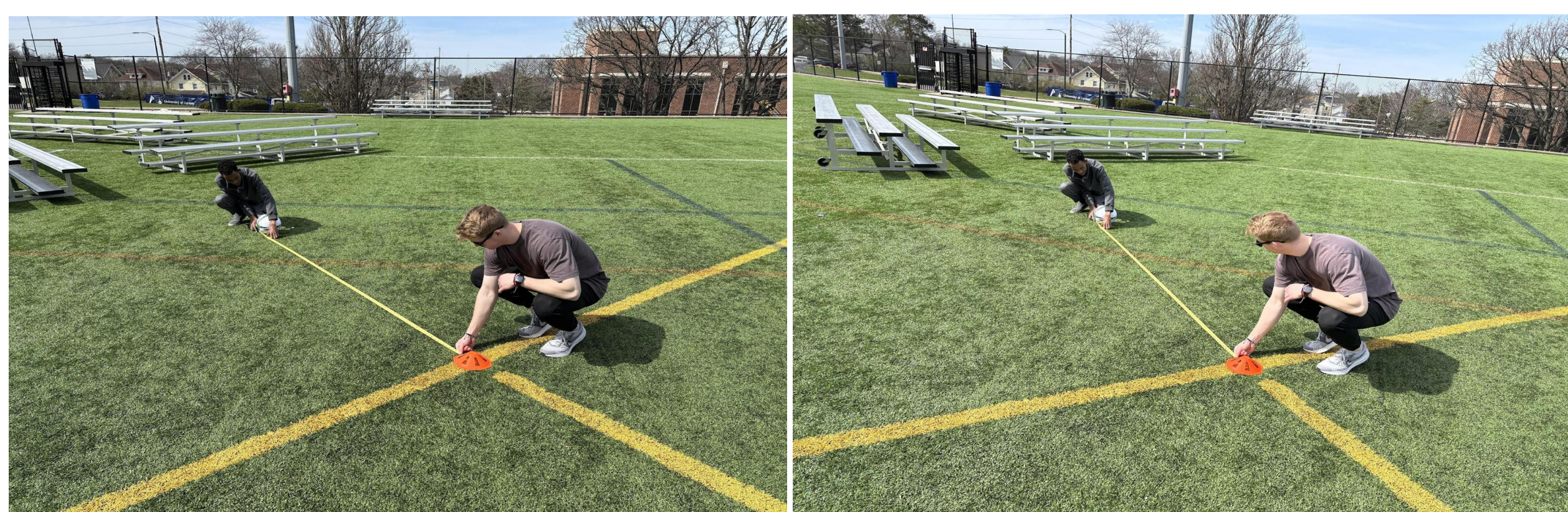
Although efforts were made to control experimental conditions, several sources of variability and potential bias may have influenced the results.

- ❖ **Learning and fatigue effects:** Although randomization and rest breaks were used, repeated kicks may still introduce improvement (learning) or decline (fatigue) over time.
- ❖ **Environmental variability:** Minor changes in wind conditions or inconsistencies in the turf surface could have affected ball movement and final position.
- ❖ **Limited blocking factor:** While kicker was included as a blocking factor, the small number of participants limits the ability to fully account for differences in individual skill levels.
- ❖ **Measurement error:** Distances were measured manually using a tape measure, which may introduce small inaccuracies in recorded values.
- ❖ **Inconsistent kicking technique:** Despite general instructions, slight differences in approach, force, or ball contact may have introduced additional variability.

Experiment Setup



Resting area



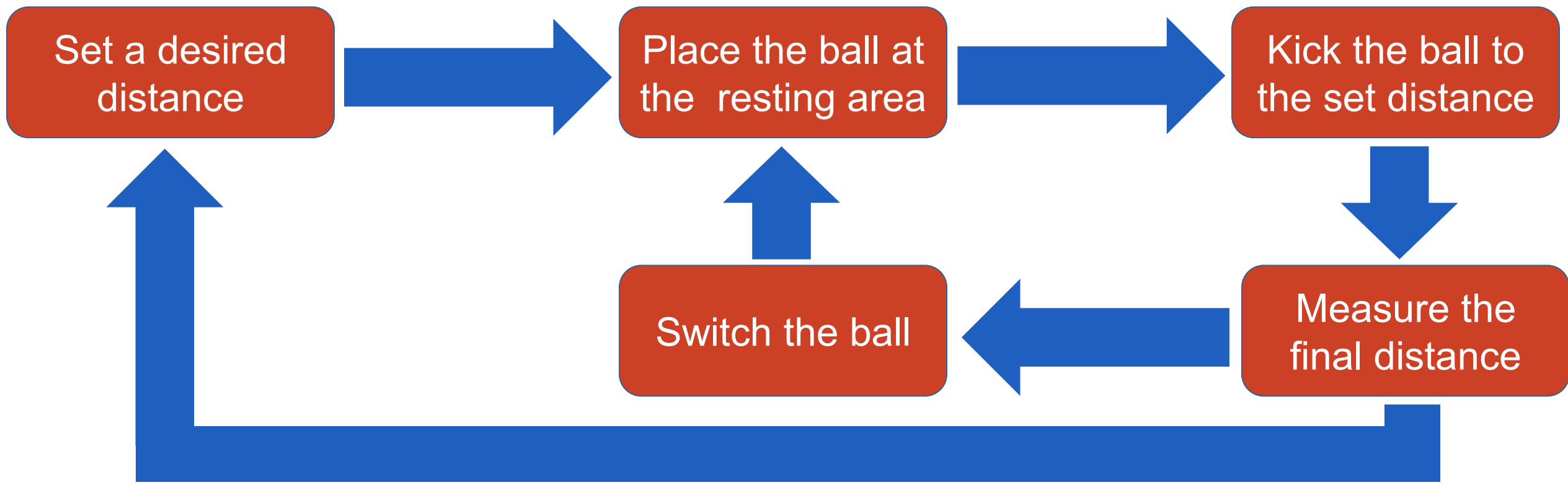
Measuring final distance



Set at 60 yards



Set at 40 yards



Discussion & Conclusions

- ❖ The interaction plot suggests non-parallel trends; however, ANOVA indicates that the interaction between ball size and starting distance is not statistically significant ($p = 0.704$).
- ❖ Starting distance is statistically significant ($p < 0.001$) and is the most influential factor, with mean error increasing noticeably from 20 to 60 yards.
- ❖ Ball size shows marginal evidence of an effect ($p \approx 0.059$), suggesting a possible but inconclusive influence that may warrant further investigation.
- ❖ The blocking factor (kicker) is not statistically significant ($p \approx 0.255$), but it is still included to account for variability between individuals.

Recommendations

- ❖ **Increase sample size:** including more participants with varying skill levels to better capture individual differences and improve the effectiveness of blocking.
- ❖ **Improve measurement precision:** use more accurate tools such as laser distance measurement to reduce potential measurement error.
- ❖ **Enhance experimental control:** Further standardize the kicking technique or provide more structured instructions to reduce variability.

Acknowledgement

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